Correlation

1 The scatter diagram shows data of the price of an item as a function of its supply. Describe the correlation as positive or negative.



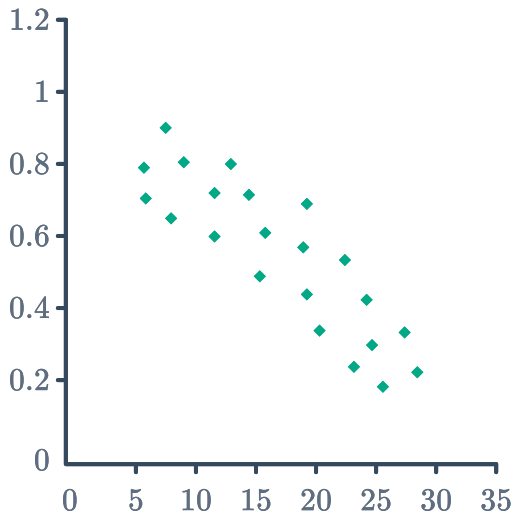
2 The scatter diagram shows data of the price of an item as a function of its demand. Describe the correlation as positive or negative.



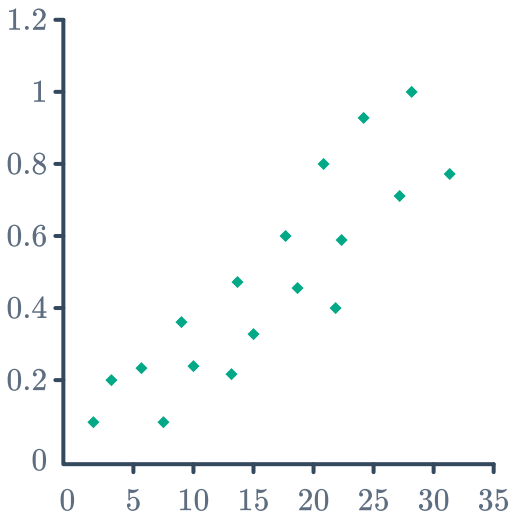
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3 For each of the following scatter diagrams, describe the correlation as positive or negative:

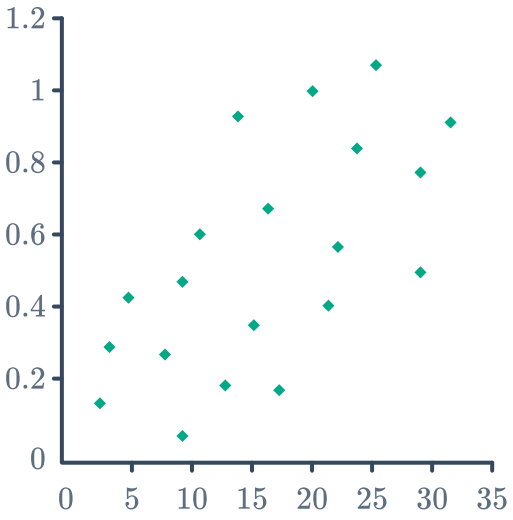
a



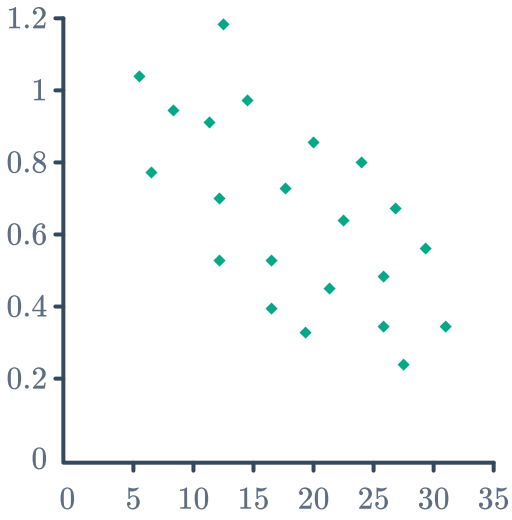
b



c



d



4 For each of the following pairs of variables:

i Is there a relationship between the two variables. If yes, answer part (ii).

ii Is the correlation between the two variables positive or negative?

a Typing speed and dancing ability.

b Temperature and ice cream sales.

c Time spent studying and exam grade.

d Eye colour and IQ.

e Temperature and soup sales.

f Rainfall and traffic accidents.

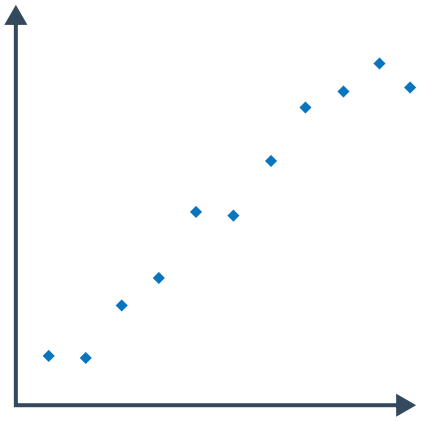
g Height and weight.

h Age and hearing ability.

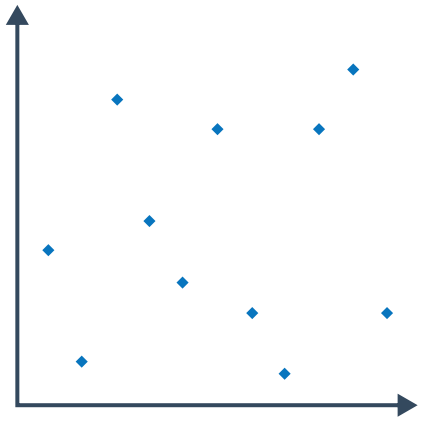
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5 For each of the following scatter diagrams, describe the correlation in terms of strength and direction:

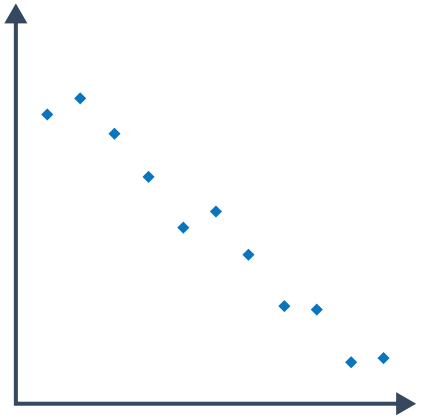
a



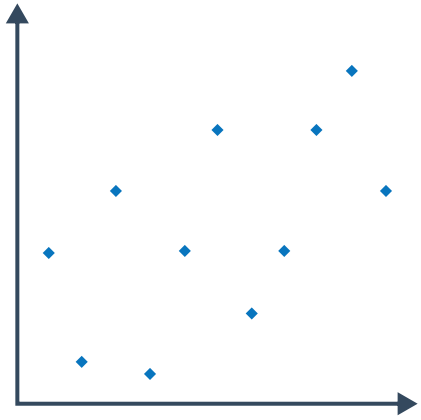
b



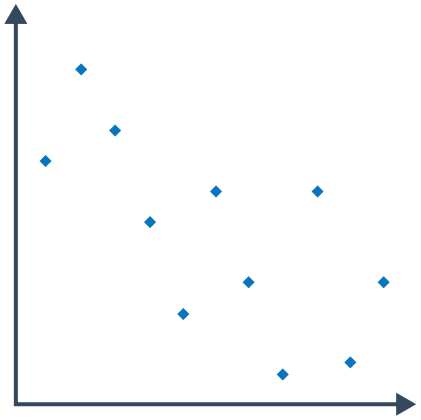
c



d



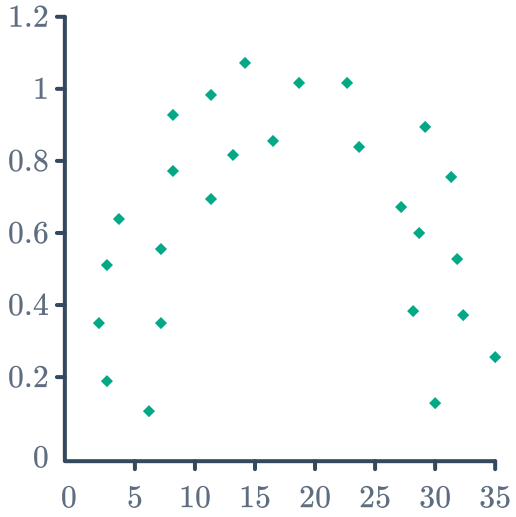
e



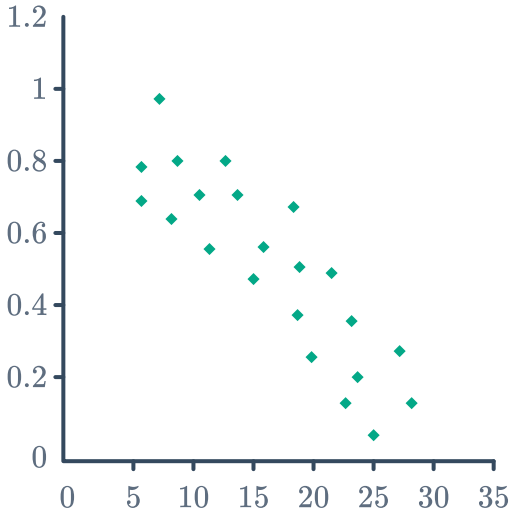
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6 For each of the following scatter diagrams, describe the data as linear or nonlinear:

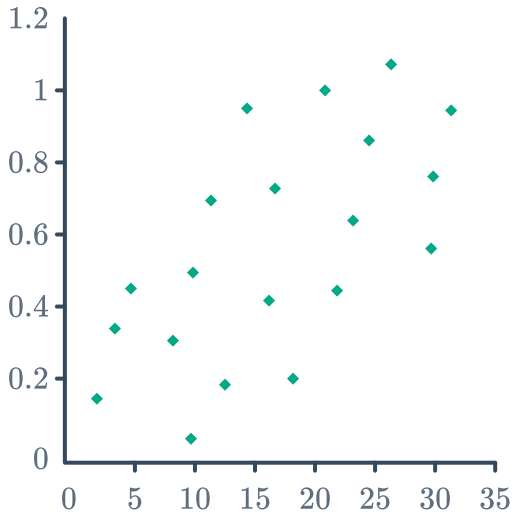
a



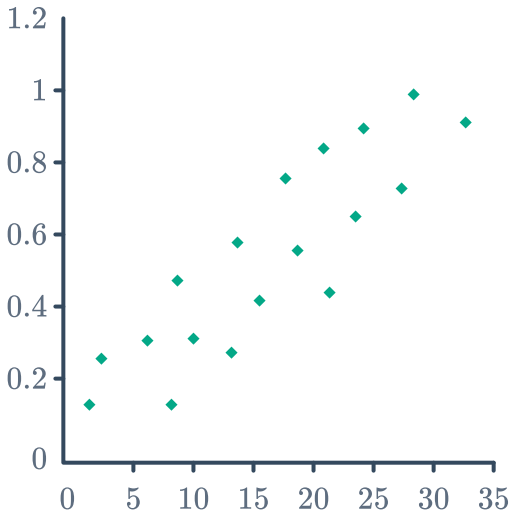
b



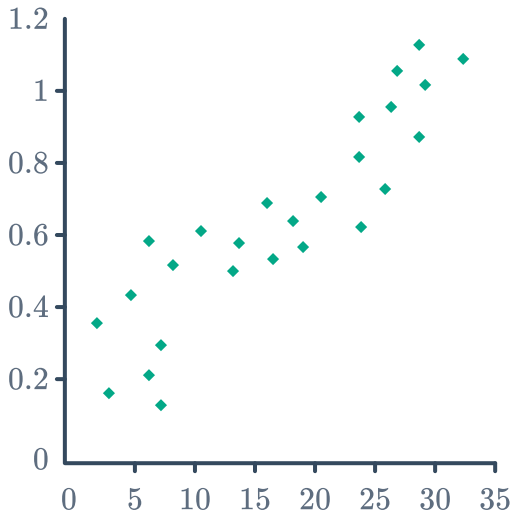
c



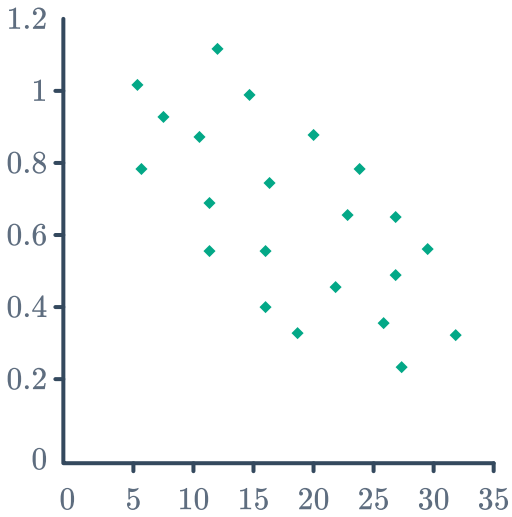
d



e



f



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7 For each of the following graphs:

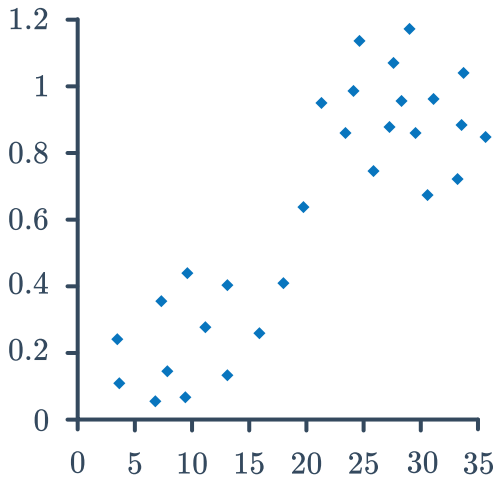
i Identify the type of correlation out of the following options:

- Linear Negative
- Linear Positive
- Nonlinear
- No Correlation

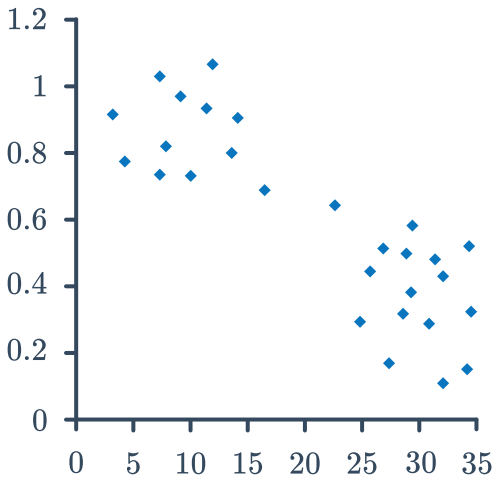
ii Identify if any of the following features are present:

- Gaps
- Clusters
- Outliers
- None

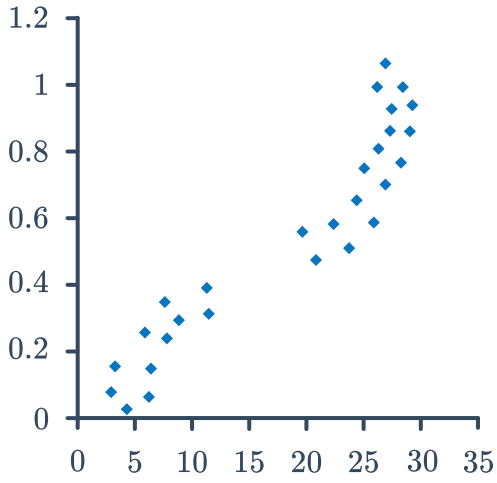
a



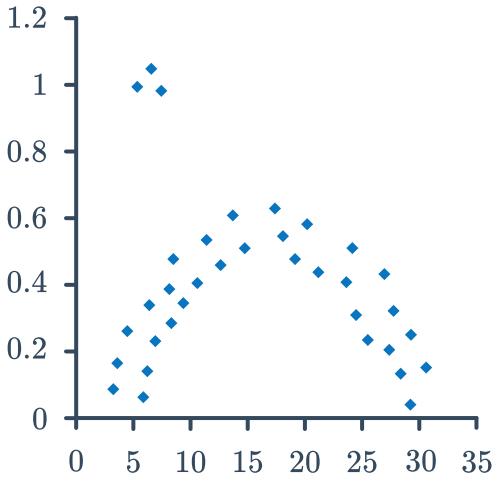
b



c

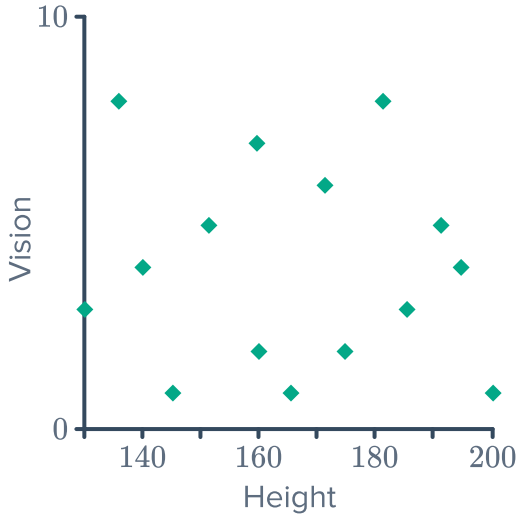


d



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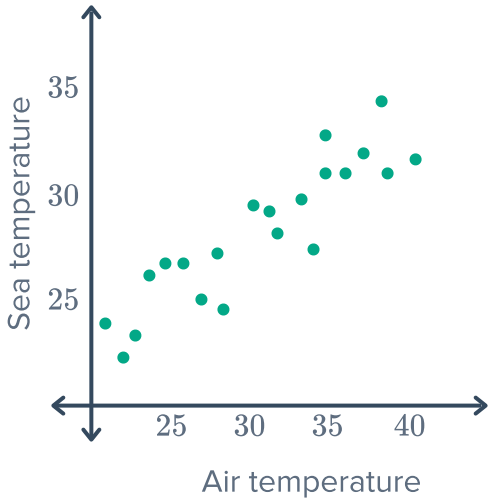
8 According to the scatter diagram shown, describe the relationship between someone's vision and their height.



9 The scatter diagram shows the relationship between air and sea temperature:

a Describe the relationship between air and sea temperature.

b Describe the correlation between air and sea temperature.

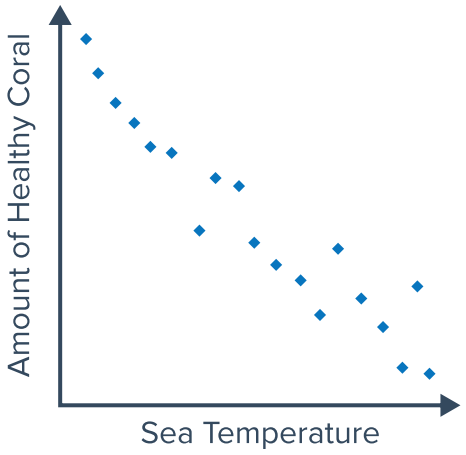


10 The following scatter diagram shows the relationship between sea temperatures and the amount of healthy coral:

a Describe the correlation between sea temperature the amount of healthy coral.

b Which variable is the dependent variable?

c Which variable is the independent variable?



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Scatter diagrams

11 A study was conducted to find the relationship between the age at which a child first speaks and their level of intelligence as teenagers. The following table shows the ages of some teenagers when they first spoke, and their results in an aptitude test:

Age when first spoke	14	27	9	9	16	21	17	10	7	19
Aptitude test results	96	69	84	90	101	87	92	99	104	93

a Construct a scatter diagram for the data.

b Describe the correlation between the data points.

12 A student was performing an experiment to study the relationship between the current and voltage through a resistor. He noted his results in the following table:

Current	1	2	3	4	5	6	7	8	9	10
Voltage	1	6	3	10	13	10	16	15	21	17

a Construct a scatter diagram for the data.

b Describe the correlation between the data points.

13 The following table displays results from an experiment:

x	1	4	5	8	9	11	13	15	18	19
y	2	4	6	8	12	24	30	46	52	64

a Construct a scatter diagram for the data.

b Describe the correlation between x and y.

14 The following table shows the time taken to finish a lap on a race track for various average speeds:

Speed	20	25	30	35	40	45	50	55	60	65
Time	85	87	75	82	69	73	60	57	45	49

a Construct a scatter diagram for the data.

b Describe the correlation between speed and time for this data.

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15 The following table shows the number of traffic accidents associated with a sample of drivers of different age groups:

Age	20	25	30	35	40	45	50	55	60	65
Accidents	41	44	39	34	30	25	22	18	19	17

a Construct a scatter diagram for the data.

b Describe the correlation between a person's age and the number of accidents.

c Does this data set contain any outliers?

16 The marks of 12 students in Maths and Sport were recorded in the following table:

a Construct a scatter diagram for the students' marks in maths vs their marks in sports.

b Describe the correlation between students' marks in maths and marks in sport.

c Which student's scores appear to represent an outlier?

Student	Mark in Maths	Mark in Sports
1	63	44
2	92	74
3	60	52
4	79	70
5	88	67
6	81	60
7	61	73
8	91	86
9	72	84
10	42	93
11	66	57
12	92	92

17 The following table shows the average IQ of a random group of people against their height:

Height (cm)	140	145	150	155	160	165	170	175	180	185
IQ	103	95	98	111	85	89	108	145	110	93

a Construct a scatter diagram for this data.

b Describe the relationship between IQ and height.

c How tall is the person who appears to be an outlier?

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18 A researcher is studying the relationship between the number of passers-by present in a situation and the time taken, in seconds, until a stranger in an emergency receives help from a passer-by. The data is recorded in the table below:

a Construct a scatter diagram to represent the data in the table.

b Comment on the relationship between the number of passers-by and the time until assistance is offered by a passer-by to a person in an emergency.

c Describe the correlation between the number of passers-by and the time until assistance is offered by a passer-by to a person in an emergency.

Passers-by	Time until offered help
1	8
2	19
3	26
4	37
5	51
6	65

19 The following table shows the marks of 12 students in English and French:

a Construct a scatter diagram for English mark vs French mark.

b Describe the correlation between students' English and French marks.

c Which student's scores appear to represent an outlier?

Student	English	French
1	85	89
2	71	71
3	57	56
4	60	62
5	79	86
6	76	75
7	71	77
8	91	86
9	50	90
10	49	47
11	66	67
12	92	92

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20 The table lists the time taken to sprint 400 metres by runners who all ran in different temperatures as part of a study:

a Construct a scatter diagram to represent the data in the table.

b How many runners were tested in the study?

c Describe the correlation between temperature and sprint time for the data.

d Which data point represents an outlier?

Temperature ($^\circ$C)	Time (sec)
5	60
2	67
10	48
8	69
1	65
7	49
6	57
4	53
3	59
9	52

21 Scientists conducted a study where each person was asked to read a paragraph and then recount as much information as they can remember. They found that the longer the paragraph, the less information each person could retain.

If the length of the paragraph were plotted (on the horizontal axis) against the amount of information retained (on the vertical axis), would the correlation be negative or positive?

22 Consider the table of values that show four excerpts from a database comparing the income per capita of a country and the child mortality rate of the country.

If a scatter diagram was created from the entire database, would you expect the relationship to be a strongly positive, strongly negative or have no relationship?

Income per capita	Child mortality rate
1465	67
11 428	16
2621	35
32 468	9

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